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Trekking Management Application

Modern Application Development 1

Project

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Project Details:

The Trekking Management Application is a web-based system developed to simplify the management of trekking activities for administrators, trek staff, and Trekkers. It provides a centralized platform to manage trek information, staff assignments, user registrations, and trek bookings efficiently.

Implementation:

- Designed and developed the Trekking Management Application using Flask and SQLite.
- Created modules for user registration, login, trek management, booking, and approval.
- Implemented separate access for Admin, Staff, and Trekkers with secure login.
- Developed a simple and responsive user interface using HTML, CSS, and Bootstrap.
- Added reports and dashboards to display booking details and application statistics.

Tools Used:

The Trekking Management Application was developed using the Flask framework with Python for the backend. SQLite was used as the database to store application data. The user interface was built using HTML, CSS, Bootstrap, and Jinja2 templates to create responsive web pages. SQLAlchemy was used for database operations, while Flask-Login managed user authentication and user

sessions. Chart.js, DataTables, and Visual Studio Code were used for generating reports, creating interactive tables, and developing the application.

DB Schema Design:

All the classes have an id field as the primary key.

1. User Class

id: user ID, integer, **primary key**
username: string, unique, cannot have null value
name: string, cannot have null value
contact: string, cannot have null value
password: string, cannot have null value
role: string (Admin, Staff, Trekker), cannot have null value
status: string (Pending, Approved, Blacklisted), cannot have null value

2. Trek Class

id: trek ID, integer, **primary key**
trek_name: string, cannot have null value
location: string, cannot have null value
difficulty: string, cannot have null value
duration: integer, cannot have null value
price: float, cannot have null value
available_slots: integer, cannot have null value
assigned_staff: **foreign key** referencing **User.id** (Staff)
status: string (Open, Full, Closed, Completed), cannot have null value

3. Booking Class (Many-to-One relationship with User and Trek)

id: booking ID, integer, **primary key**
user_id: **foreign key** referencing **User.id**
trek_id: **foreign key** referencing **Trek.id**
booking_date: date, cannot have null value
participants: integer, cannot have null value
status: string (Booked, Cancelled), cannot have null value

Relation between Classes

User.id → **Trek.assigned_staff** (Foreign Key)
User.id → **Booking.user_id** (Foreign Key)
Trek.id → **Booking.trek_id** (Foreign Key)

ER Diagram: [LINK](#)

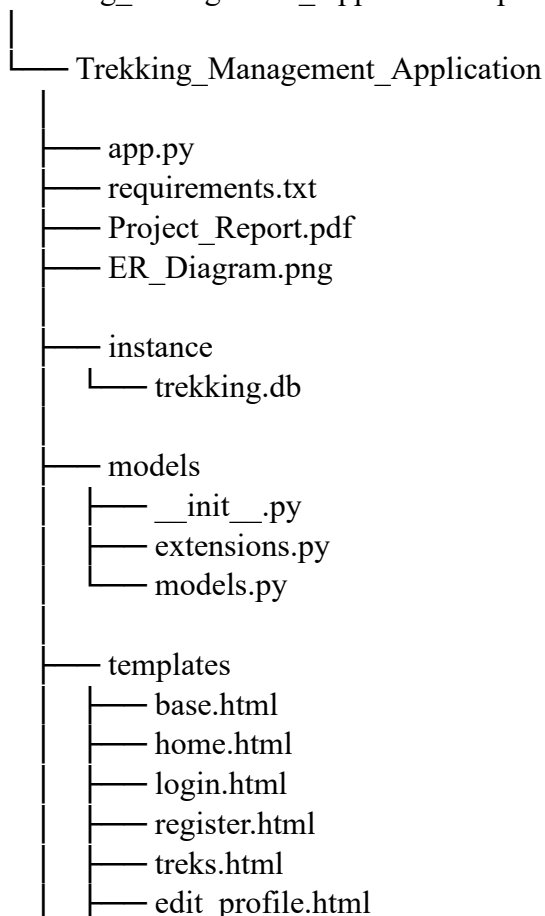
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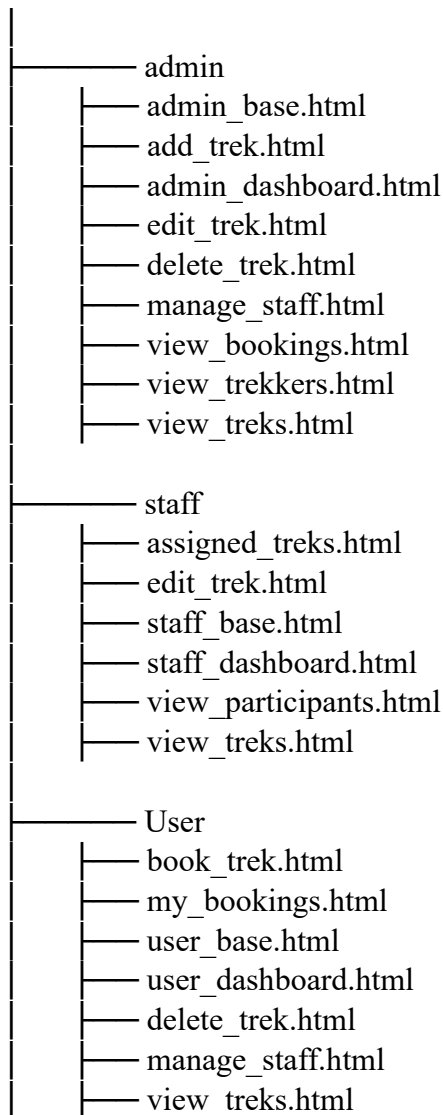


Folder Structure: [LINK](#)

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Trekking_Management_Application.zip





Workflow:

The application starts at **<http://127.0.0.1:5000>**, which displays the Home, Treks, Login, and Register pages.

- * Home is a simple welcome page.
- * Treks allows anyone to view all trek details without any credentials.
- * A common Login page is provided for all users.
- * The Register page is used by Trekkers and Trek Staff.

The Admin account is created automatically when `app.py` is run, so no separate registration is required for the administrator.

Users can register as either a Trekker or Trek Staff. If the user registers as Staff, the Administrator must approve the account before login. If the user registers as a Trekker, the account is approved automatically.

All users use the same login page. Based on the role stored in the database, users are redirected to their respective dashboards (Admin, Staff, or Trekker).

The Administrator dashboard provides Home, Staff, Trekkers, Add Trek, and View Treks options. The Administrator manages users, approves staff registrations, creates treks, assigns staff, and monitors bookings.

Staff members can view their assigned treks, update trek details, and view the list of participants.

Trekkers can browse available treks, book a trek, view their bookings, and cancel bookings if required.

All user, trek, and booking information is stored in the SQLite database and is updated automatically whenever changes are made.

The dashboard displays summaries such as the total number of treks, users, staff, and bookings using charts and tables.

Conclusion:

The Trekking Management Application provides an efficient platform for managing trekking activities. It simplifies user registration, trek management, staff assignment, and booking operations through a role-based system. The application offers a user-friendly interface and helps administrators, staff, and trekkers perform their tasks efficiently.

Presentation Video [Link](https://drive.google.com/file/d/1jGdZg2Vr7tpS0DfTjBMV-m8kYBObP03c/view?usp=sharing)

<https://drive.google.com/file/d/1jGdZg2Vr7tpS0DfTjBMV-m8kYBObP03c/view?usp=sharing>